

MTH209: Assignment 10

Write a function that takes a dataset, and calculates the total within cluster sums of squares of a k -means implementation for $k = 1, \dots, \text{max.k}$

The total within cluster sums of squares is the `tot.withinss` component of the `kmeans` output object.

Function format must return a matrix of 2 columns according to the format below.

```
# data: is fed directly to k-means
# max.k: integer
track_within_ss <- function(data, max.k)
{
  # first column is k and second column is total within cluster ss
  output <- matrix(0, ncol = 2, nrow = max.k)
  output[,1] <- 1:max.k
  ...
  ...

  return(output)
}
```

NOTE: Only upload the function in `assignment10.R` and nothing else.